That concludes the forecasting project explanation. Next, we will look at the DL project.

Project: Multilabel Image Classification

This project is a DL data solution that uses another Kaggle dataset. We will use these datasets images to fine-tune a deep-learning model, using PyTorch and Lightning to predict a corresponding label. We will implement MLflow code for experiment and model tracking, Spark for fast training and **inference**, and Delta for data version control. We will deploy the model as we would for a real-time scenario by creating a model wrapper, similar to the wrapper we use for the streaming transactions project. Take a look through the project pipeline in *Figure 1.9* to understand how we'll progress through this project.

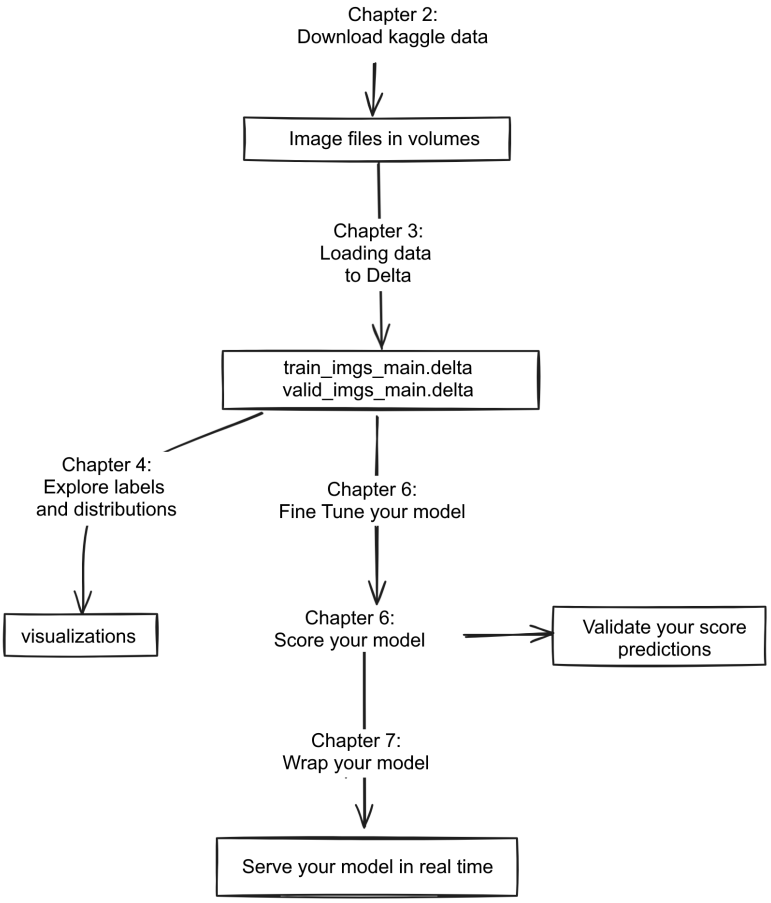


Figure 1.9 – The project pipeline for the multilabel image classification project

That concludes the image classification project explanation. Next, we will look at the chatbot project.